Terraform Challenge - 02

- Write a code using Terraform that deploys and achieves the following:-
  Create an EC2 Instance on AWS Cloud using the below-mentioned ubuntu AMI and instance_type or, (You can use your choice as well) ami = "ami-0461b1436c9b3c1a3" instance_type = "t2.micro"
  Install a web server on the same EC2 Instance. The web server should listen on port 8080.

- Create a Docker-compose file for the project: https://github.com/betawins/cricbuzz-docker-hello-world.git and point it to port 8000.
- Make the web server point to the above service so that 8080 returns the response from the Docker service.
- Configure to Save the Logs in the location /var/log/cb.log
- Write the Log rotate configuration for the above file
- modules > ec2 > main.tf > resource "aws_instance" "cb_server"

```tf
resource "aws_instance" "cb_server" {

  ami           = var.ami
  instance_type = var.instance_type
  key_name = "Linux-key"

  security_groups = [var.security_group]

  user_data = file("${path.module}/userdata.sh")

  tags = {
    Name = "cb-server"
  }

}
```

- modules > ec2 > variables.tf > variable "security_group"

```tf
variable "ami" {

}

variable "instance_type" {

}

variable "security_group" {

}
```

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modules > security_group > main.tf > resource "aws_security_group" "cb_sg"

```
resource "aws_security_group" "cb_sg" {

  name = "cb-security-group"

  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port   = 8080
    to_port     = 8080
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port   = 8000
    to_port     = 8000
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

}
```

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modules > ec2 > $ userdata.sh

```bash
#!/bin/bash

# update system
dnf update -y

# install required packages
dnf install -y docker nginx git logrotate

# install docker compose
curl -L https://github.com/docker/compose/releases/latest/download/docker-compose-linux-x86_64 -o /usr/local/bin/docker-compose
chmod +x /usr/local/bin/docker-compose

# start and enable docker
systemctl start docker
systemctl enable docker

# wait for docker to fully start
sleep 20

# start and enable nginx
systemctl start nginx
systemctl enable nginx

# clone repo
cd /home/ec2-user
git clone https://github.com/betawins/cricbuzz-docker-hello-world.git

cd cricbuzz-docker-hello-world

# create docker compose file
cat <<EOF > docker-compose.yml
version: "3"
services:
  hello-world:
    image: ghcr.io/infrastructure-as-code/hello-world
    ports:
      - "8000:8080"
EOF

# run container
docker-compose up -d

# nginx configuration
cat <<EOF > /etc/nginx/conf.d/cb.conf
server {
    listen 8080;

    location / {
        proxy_pass http://localhost:8000;
    }

    access_log /var/log/cb.log;
}
EOF

# restart nginx
systemctl restart nginx

# configure logrotate
cat <<EOF > /etc/logrotate.d/cb
/var/log/cb.log {
    daily
    rotate 7
    compress
    missingok
    notifempty
}
```

-

modules > security_group > output.tf > output "sg_name"

```tf
output "sg_name" {
  value = aws_security_group.cb_sg.name
}
```

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- main.tf > module "ec2"

```
module "security_group" {
  source = "./modules/security_group"
}

module "ec2" {
  source = "./modules/ec2"

  ami           = var.ami
  instance_type = var.instance_type
  security_group = module.security_group.sg_name
}
```

- provider.tf > provider "aws"

```
provider "aws" {
  region = "us-east-1"
}
```

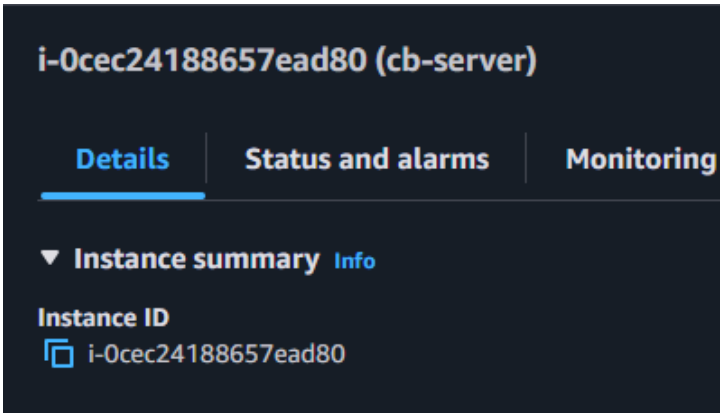
- variables.tf > variable "ami"

```
variable "ami" {
  default = "ami-0f4b77a23eac2378a"
}

variable "instance_type" {
  default = "t2.micro"
}
```

-

```
module.security_group.aws_security_group.cb_sg: Creating...
module.security_group.aws_security_group.cb_sg: Creating...
module.security_group.aws_security_group.cb_sg: Creation complete after 5s [id=sg-0517dcd62e5dccf2e]
module.ec2.aws_instance.cb_server: Creating...
module.ec2.aws_instance.cb_server: Still creating... [00m10s elapsed]
module.ec2.aws_instance.cb_server: Creation complete after 16s [id=i-0cec24188657ead80]
module.ec2.aws_instance.cb_server: Creation complete after 16s [id=i-0cec24188657ead80]


Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
```

-



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- sg-0517dcd62e5dccf2e - cb-security-group

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (3)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
–	sgr-05797095edc3ff29a	IPv4	Custom TCP	TCP	8000	0.0.0.0/0
–	sgr-0fbfe19ddef7c8737	IPv4	SSH	TCP	22	0.0.0.0/0
–	sgr-087a72b58e781eafd	IPv4	Custom TCP	TCP	8080	0.0.0.0/0

-

```
STATUS          PORTS                                        NAMES
Up 12 minutes   0.0.0.0:8000->8080/tcp, :::8000->8080/tcp    cricbuzz-docker-hello-world-hello-world-1
```

-

```
[root@ip-172-31-29-158 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-03-09 09:11:09 UTC; 12min ago
    Process: 5939 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 5940 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 5941 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 5942 (nginx)
      Tasks: 2 (limit: 1120)
     Memory: 2.5M
        CPU: 41ms
```

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Not secure 54.242.134.135:8080

Watch Cartoons an... | Minimalism wallpap... | YouTu

```
Hello, World!
```

-

```
[root@ip-172-31-29-158 ~]# cat /var/log/cb.log
124.123.170.76 - - [09/Mar/2026:09:13:15 +0000] "GET / HTTP/1.1" 200 13 "-" "Mozilla/5.0 (Wi
124.123.170.76 - - [09/Mar/2026:09:13:16 +0000] "GET /favicon.ico HTTP/1.1" 404 18 "http://5
[root@ip-172-31-29-158 ~]#
```

-

```
[root@ip-172-31-29-158 ~]# ls /home/ec2-user
cricbuzz-docker-hello-world
[root@ip-172-31-29-158 ~]# cat /etc/logrotate.d/cb
/var/log/cb.log {
    daily
    rotate 7
    compress
    missingok
    notifempty
}
[root@ip-172-31-29-158 ~]#
```

-